Lecture 25 10/11/2026

# Taylor Applications

Near the point $x_0$, the Taylor polynomial of the function $f(x)$ is

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!}f''(x_0)(x-x_0)^2 + \dots$$

$$\dots + \frac{1}{n!}f^{(n)}(x_0)(x-x_0)^n + R_n(x).$$

The most important element of this series is not the entire polynomial but the **first non-zero term**: generally this term determines the sign, the scale and the shape of the function $f(x)$ close to $x_0$.

## PEANO FORM (for the remainder)

In the previous lecture we used the **Lagrange remainder form**.

Sometimes we use the **Peano form**

The Taylor's formula can be rewritten as

$$f(x) = P_n(x) + o\left((x-x_0)^n\right), \quad x \to x_0.$$

↳ the remainder is $r(x) = o\left((x-x_0)^n\right)$.

This means that $\dfrac{r(x)}{(x-x_0)^n} \xrightarrow{x \to x_0} 0$

"The remainder becomes negligible compared with $(x-x_0)^n$ when $x \to x_0$"

The MacLaurin expansion can be rewritten in the Peano form

As $x \to 0$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

ORDER OF A ZERO

Suppose that

$$f(x_0) = f'(x_0) = f''(x_0) = \dots = f^{(m-1)}(x_0) = 0$$

and $f^{(m)}(x_0) \neq 0$.

Then

$$f(x) = \frac{f^{(m)}(x_0)}{m!}(x-x_0)^m + o((x-x_0)^m)$$

and therefore $f(x) \sim \dfrac{f^{(m)}(x_0)}{m!}(x-x_0)^m$, $x \to x_0$.

This means that the local behaviour of $f(x)$ near to $x_0$ depends only on the $m$-th derivative $f^{(m)}(x)$ in $x_0$.

EX: $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$

First of all we expand with Taylor the function $e^x$ close to $x_0 = 0$:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3).$$

Then we rewrite the numerator moving the terms 1+x in the left side

$$e^x = (1+x) + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$\hookrightarrow e^x - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

Now we compute the limit gathering the term $x^2$:

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x^2} = \lim_{x\to 0} \frac{\frac{x^2}{2} + \frac{x^3}{6} + o(x^3)}{x^2}$$

$$= \lim_{x\to 0} \frac{x^2}{x^2} \cdot \frac{\frac{1}{2} + \frac{x}{6} + o(x)}{1}$$

$$\rightsquigarrow = o(x^3)\cdot\frac{1}{x^2}$$

$$= o\left(\frac{x^3}{x^2}\right)$$

$$= o(x)$$

$$= \lim_{x\to 0} \left( \frac{1}{2} + \frac{x}{6} + o(x) \right) = \frac{1}{2}.$$

**EX** $\lim_{x\to 0} \dfrac{\sin(x) - x}{x^3}$

We expand $\sin(x)$ around $x_0 = 0$.

$$\sin(x) = x - \frac{x^3}{6} + o(x^3)$$

Therefore $\sin(x) - x = -\dfrac{x^3}{6} + o(x^3)$

$$\lim_{x\to 0} \frac{\sin(x) - x}{x^3} = \lim_{x\to 0} \frac{-\frac{x^3}{6} + o(x^3)}{x^3} = \lim_{x\to 0} \frac{x^3}{x^3} \frac{-\frac{1}{6} + o(1)}{1}$$

$$= -\frac{1}{6}$$

**EX** $\lim_{x\to 0} \dfrac{\ln(1+x) - x}{x^2}$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3) \quad \to \quad \ln(1+x) - x = -\frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

$$\lim_{x\to 0} \frac{\ln(1+x) - x}{x^2} = \lim_{x\to 0} \frac{-\frac{x^2}{2} + \frac{x^3}{3} + o(x^3)}{x^2} = \lim_{x\to 0} \frac{x^2}{x^2} \frac{-\frac{1}{2} + \frac{x}{3} + o(x)}{1}$$

$$= \lim_{x\to 0} \left( -\frac{1}{2} + \frac{x}{3} + o(x) \right) = -\frac{1}{2}$$

## How many Taylor terms do we need?

It depends on the problem. If you are comparing a function with a polynomial of order $n$, you have to expand in Taylor your function at the order $n$.

$$\lim_{x\to 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3) \Rightarrow e^x - 1 - x - \frac{x^2}{2} = \frac{x^3}{6} + o(x^3)$$

$$\lim_{x\to 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^3} = \lim_{x\to 0} \frac{\frac{x^3}{6} + o(x^3)}{x^3} = \lim_{x\to 0} \frac{x^3}{x^3} \cdot \frac{\frac{1}{6} + o(1)}{1}$$

$$= 1/6$$

## POWER COMPARISON

Suppose $f(x) \sim c(x-x_0)^m$, $g(x) \sim d(x-x_0)^n$, $c, d \neq 0$, $x \to x_0$.

Then $$\frac{f(x)}{g(x)} \sim \frac{c}{d}(x-x_0)^{m-n}.$$

If $m > n$ ratio $f(x)/g(x) \to \infty$

If $m = n$ ratio $f(x)/g(x) \to c/d$

If $m < n$ ratio $f(x)/g(x) \to \infty$

EX: $\lim_{x\to 0} \dfrac{e^x - 1 - x}{1-\cos x}$. Solve this limit using

$e^x - 1 - x \sim \dfrac{x^2}{2}$ and $1-\cos(x) \sim \dfrac{x^2}{2}$

$$\frac{e^x - 1 - x}{1-\cos(x)} \sim \frac{x^2/2}{x^2/2} = \frac{x^2}{x^2}\,\frac{1/2}{1/2} = \frac{1/2}{1/2} = 1.$$

↳ $\lim_{x\to 0} \dfrac{e^x - 1 - x}{1-\cos(x)} = 1.$

You can use also De l'Hôpital because you have a $\frac{0}{0}$ form.

## LOCAL SIGN

Suppose a function $f(x)$ such that it is represented via the first non-zero Taylor term:

$$f(x) = c(x-x_0)^m + o\left((x-x_0)^m\right), \quad c \neq 0.$$

Close to $x_0$, the sign of $f(x)$ is the sign of $c(x-x_0)^m$.

1) If $m$ is even, the sign is the same on both sides of $x_0$

2) If $m$ is odd, the sign changes across $x_0$,

↳ The sign of $c$ determines the sign around $x_0$

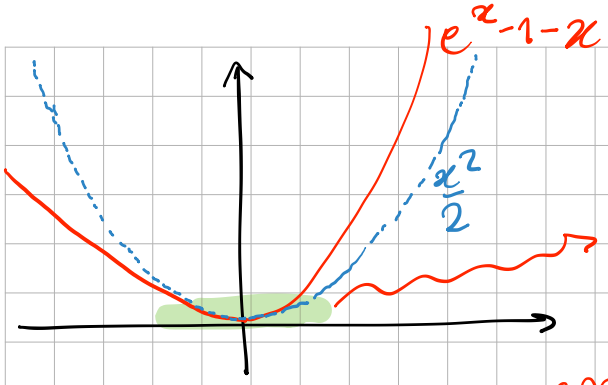
EX: We want to study the sign of the function $e^x - 1 - x$ near the origin.

Using Taylor we get $e^x - 1 - x = \dfrac{x^2}{2} + o(x^2)$

→ The first non-zero term $x^2$ has an even order this means that the sign is equal before and after the point $x_0 = 0$.

→ The coefficient of the first non-zero term is $1/2 > 0$, positive.

The consequence is that $e^x - 1 - x > 0$ for small $x \neq x_0 = 0$.




in this region, for small $x \neq x_0 = 0$, the function $e^x - 1 - x$ is well approximated by $\frac{x^2}{2}$.

EX: We study $\frac{\sin(x)}{x}$ near the origin

Using Taylor $\sin(x) = x - \frac{x^3}{6} + o(x^3)$

$\hookrightarrow \sin(x) - x = -\frac{x^3}{6} + o(x^3)$

→ The first non-zero term $x^3$ has an odd order. This means that before and after the point $x_0$ we'll see a change in the sign.

in particular close to zero, $x^3 \geq 0$ for $x > 0$

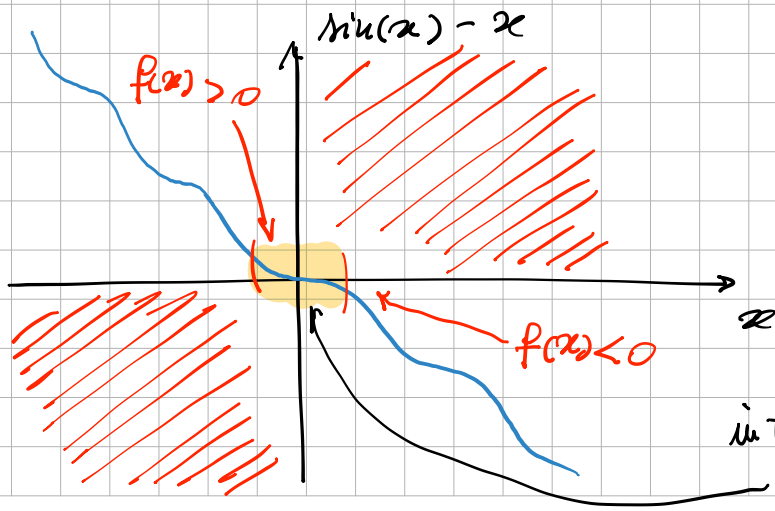
$x^3 < 0$ for $x < 0$

→ The coefficient of the first non-zero term is $-\frac{1}{6} < 0$.

Therefore we have to change the sign, close to zero:

$-\frac{1}{6}x^3 < 0$ for $x > 0$

$-\frac{1}{6}x^3 > 0$ for $x < 0$




**Pay attention**

Using Taylor we have information only near the point of expansion, in this case the origin

# LOCAL ETREMA

Assume $f'(x_0) = 0$. Now we take in account the first derivate after $f'(x)$ which doesn't vanish in $x_0$. For instance this is $f^{(m)}(x_0)$. Then locally

$$f(x) - f(x_0) \sim \frac{f^{(m)}(x_0)}{m!}(x-x_0)^m$$

this means that they are asymptotically equivalent when $x \to x_0$. So in this case we don't want to establish a perfect equality. For this reason we don't need to write $o((x-x_0)^m)$.

$o((x-x_0)^m)$ is the remainder, the error. We use it if and only if we write a perfect equality.

- If $m$ is even: → positive coefficient → local minimum
  → negative coefficient → local maximum
- If $m$ is odd: no local extremum.

EX: At $x_0 = 0$ $f(x) = x^4$ has

$$f'(0) = 4x^3\big|_{x=0} = 0$$
$$f''(0) = 12x^2\big|_{x=0} = 0$$
$$f'''(0) = 24x\big|_{x=0} = 0$$
$$f^{(4)}(0) = 24 > 0.$$

zero is a local minimum ←

At the same point $f(x) = -x^4$ has

$$f'(0) = -4x^3\big|_{x=0} = 0$$
$$f''(0) = -12x^2\big|_{x=0} = 0$$
$$f'''(0) = -24x\big|_{x=0} = 0$$

$f^{(4)} = -24 < 0 \to$ local maximum in $x_0 = 0$.

# LOCAL BEHAVIOUR AT A NON-ZERO POINT

Consider $f(x) = \ln(x) - x + 1$ near to $x_0 = 1$.

To simplify the function we set $t = x - 1$

$t = x - 1 \longrightarrow x = t + 1$ and now we substitute $t+1$ in $x$:

$$f(x) = \ln(x) - x + 1 \longrightarrow f(t) = \ln(t+1) - (t+1) + 1$$
$$= \ln(1+t) - t$$

Using Taylor we know that

$$\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} + o(t^3)$$

$$\hookrightarrow f(t) = \left(t - \frac{t^2}{2} + \frac{t^3}{3} + o(t^3)\right) - t$$

$$= -\frac{t^2}{2} + \frac{t^3}{3} + o(t^3)$$

$$= -\frac{t^2}{2} + o(t^2)$$

we gather all the the terms of the third order. In this way we are making an error of the second order.

$t = x - 1 \longrightarrow$

$$= -\frac{(x-1)^2}{2} + o\left((x-1)^2\right)$$

Therefore $\ln(x)$ in $x_0 = 1$ is null ($f(1) = 0$), but the order is 2 so we have a local extrema, and the coefficient is negative $\Rightarrow$ local maximum.

For the sign: the order is even, so the same behaviour before and after $x_0 = 1$ but with the negative sign.

# Function study

$f(x) = \ln(x) - x + 1$

**Domain:** $D = (0, +\infty)$ because we have to impose $x > 0$.

**Intercepts:**

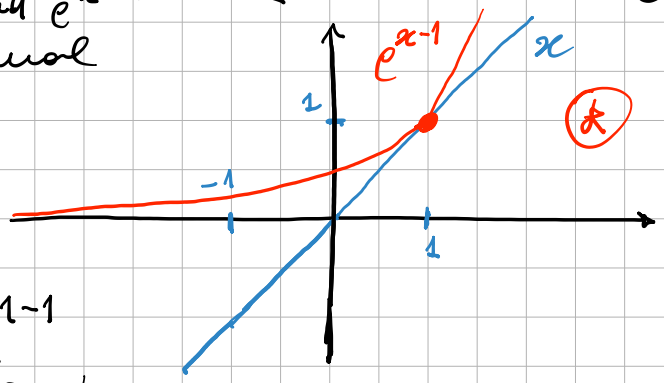
x-axis $\Rightarrow \begin{cases} f(x) = \ln(x) - x + 1 \\ y = 0 \end{cases} \rightarrow \ln(x) - x + 1 = 0$

$\ln(x) = x - 1$

$e^{\ln(x)} = e^{x-1}$

$x = e^{x-1}$

here we are studying when $x$ and $e^{x-1}$ are equal




in $x = 1$

$x = e^{x-1} \Leftrightarrow 1 = e^{1-1}$

$1 = e^0$

$1 = 1$ ✓

$f(1) = \ln(1) - 1 + 1 = 0 - 1 + 1 = 0$

So the point $P: (1, 0)$ is an intersection with the x-axis.

y-axis: $x = 0$. The function is not defined for $x = 0$, so we haven't intersection with the y-axis.

**Sign** $f(x) > 0 \rightarrow \ln(x) - x + 1 > 0$

$\ln(x) > x - 1$

$e^{\ln(x)} > e^{x-1}$

$x > e^{x-1}$ Never! Observe the graph (*)

here the function is null

$+ \quad 0 \quad - \quad 1 \quad + \quad - \quad - \quad -$

$f(x) < 0$ in $(0, 1) \cup (1, +\infty)$.

# limits

We have to study the limit near to extremes of our domain:

$$\lim_{x \to 0^+} f(x) \quad \text{and} \quad \lim_{x \to +\infty} f(x)$$

$$\lim_{x \to 0^+} \left( \ln(x) - x + 1 \right) = -\infty$$

$$\lim_{x \to +\infty} \left( \ln(x) - x + 1 \right) = +\infty - \infty \; !$$

$$\lim_{x \to +\infty} \left( \ln(x) - x + 1 \right) = \lim_{x \to +\infty} x \left( \frac{\ln(x)}{x} - 1 + \frac{1}{x} \right) =$$

$$= \lim_{x \to \infty} (x) \cdot \lim_{x \to +\infty} \left( \frac{\ln(x)}{x} - 1 + \frac{1}{x} \right) =$$

$\frac{\ln(x)}{x} \to 0$, $\frac{1}{x} \to 0$

$\frac{\infty}{\infty}$ we use de l'Hôpital

$$\lim_{x \to \infty} \frac{\ln(x)}{x} = \lim_{x \to \infty} \frac{1/x}{1} = \lim_{x \to \infty} \frac{1}{x} = 0$$

$$= \lim_{x \to \infty} (x) \cdot \lim_{x \to \infty} \left( \frac{\ln(x)}{x} - 1 + \frac{1}{x} \right) = \lim_{x \to \infty} (-1)(x) = -\infty$$

$\frac{\ln(x)}{x} \to 0$, $\frac{1}{x} \to 0$

$= (-1)$

**First derivative:** $\frac{d}{dx} \left( \ln(x) - x + 1 \right) = \frac{1}{x} - 1$

To find the critical points we set $f'(x) = 0$

$$\frac{1}{x} - 1 = 0 \rightarrow \frac{1}{x} = 1$$

The unique solution is for $x_0 = 1$ : $\frac{1}{1} = 1$

therefore $x_0 = 1$ is a critical point.

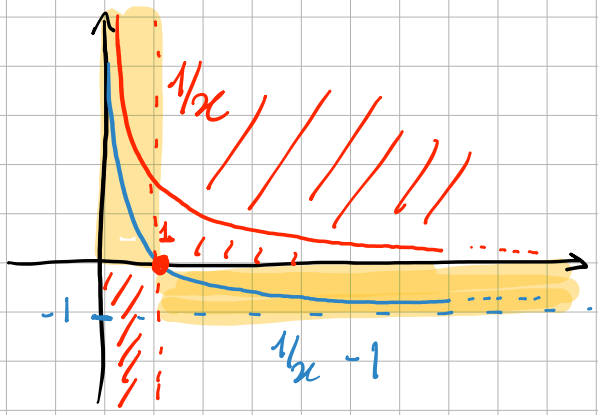
$f(x_0) = f(1) = 0$. $P: (1, 0)$ is an intersection with the x-axis and the same time a critical point.

To understand the nature of $x_0$, we study the sign of $f'(x)$.

$f(x)$ is defined in $(0, +\infty)$, so we are interested only in this domain

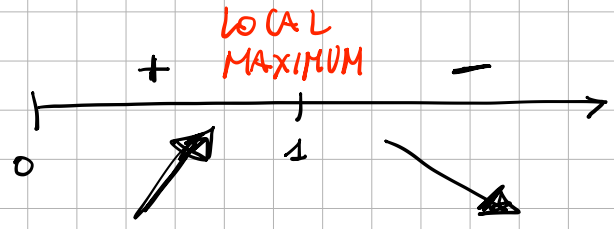
$f'(x) > 0$

$\frac{1}{x} - 1 > 0 \rightarrow \frac{1}{x} > 1 \rightarrow x < \frac{1}{1} = 1 \Rightarrow x < 1$




$f'(x) > 0$, $x < 1$ or $x \in (0,1)$

$f'(x) < 0$, $x > 1$ or $x \in (1, +\infty)$

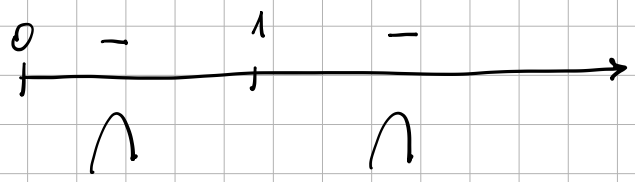



## Second derivative

$$\frac{d}{dx}\left(\frac{d}{dx} f(x)\right) = \frac{d^2}{dx^2} f(x) = f''(x) = \frac{d}{dx}\left(\frac{1}{x} - 1\right) = -\frac{1}{x^2}.$$

$-\frac{1}{x^2} < 0 \quad \forall x \in (0, +\infty)$. We have zero point such that $f''(x) = 0$.

So zero inflection points.



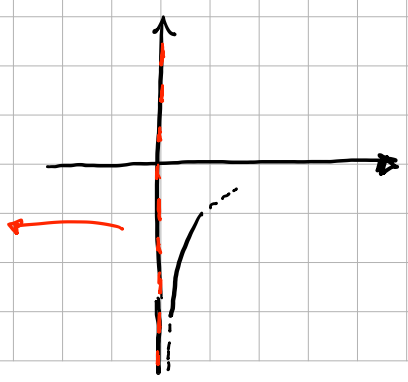

$f''(x) < 0$ for $x \in (0,1) \cup (1, +\infty)$

$f(x)$ is concave in $(0,1) \cup (1, +\infty)$

## Asymptotes

We have a vertical asymptote because we get an infinite point in the limit $x \to 0^+$. The vertical asymptote is represented by

$x = 0$.



the red dashed line represents the vertical asymptote.

We have zero horizontal asymptotes because we get zero finite limit for $x \to \pm\infty$. Remember that we get an horizontal asymptote $y = L$ if and only if

$$\lim_{x \to -\infty} f(x) = L \quad \text{or} \quad \lim_{x \to +\infty} f(x) = L$$

when $L$ is a finite number: $|L| < +\infty$.

To understand if we have an oblique asymptote, we study

$$m = \lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{\ln(x) - x + 1}{x} = \lim_{x \to \infty} \frac{\ln(x)}{x} + \lim_{x \to \infty} \left(-\frac{x+1}{x}\right) = -1$$

($\frac{\ln(x)}{x} \to 0$, $\frac{x+1}{x} \to -1$)

Now we study $q = \lim_{x \to +\infty} (f(x) - mx) \overset{m=-1}{=} \lim_{x \to \infty} (f(x) + x)$

$$= \lim_{x \to \infty} (\ln(x) - x + 1 + x) = \lim_{x \to \infty} (\ln(x) + 1) = +\infty$$

$q$ is infinite $\Rightarrow$ the oblique asymptote doesn't exist, because a line $y = mx + q$ such that $\lim_{x \to \infty} [f(x) - mx - q] = 0$ doesn't exist.

**Continuity** The function is continuous everywhere in the domain $(0, +\infty)$ because it is a linear combination of continuous function: $\ln(x)$, $-x$, $+1$.